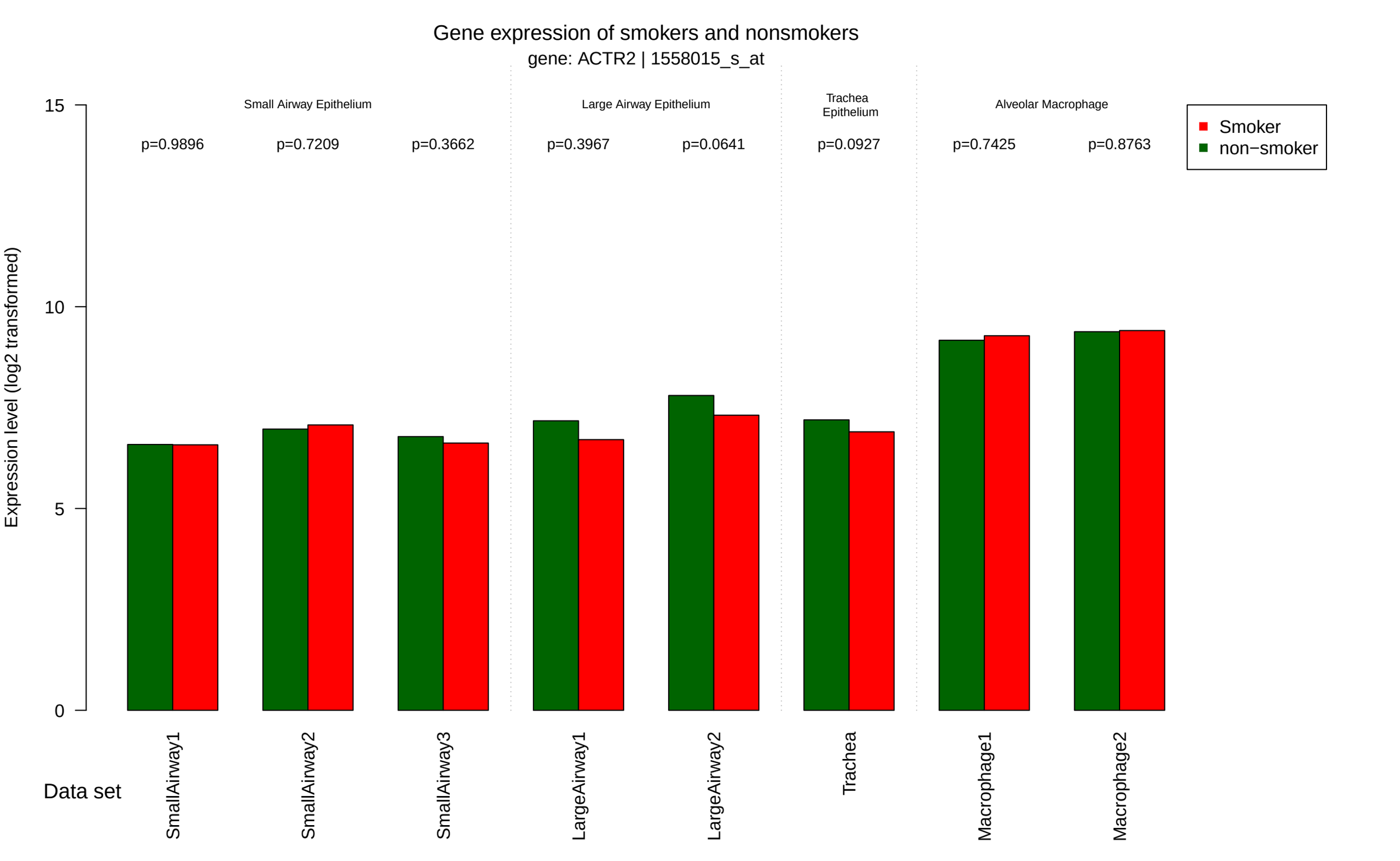
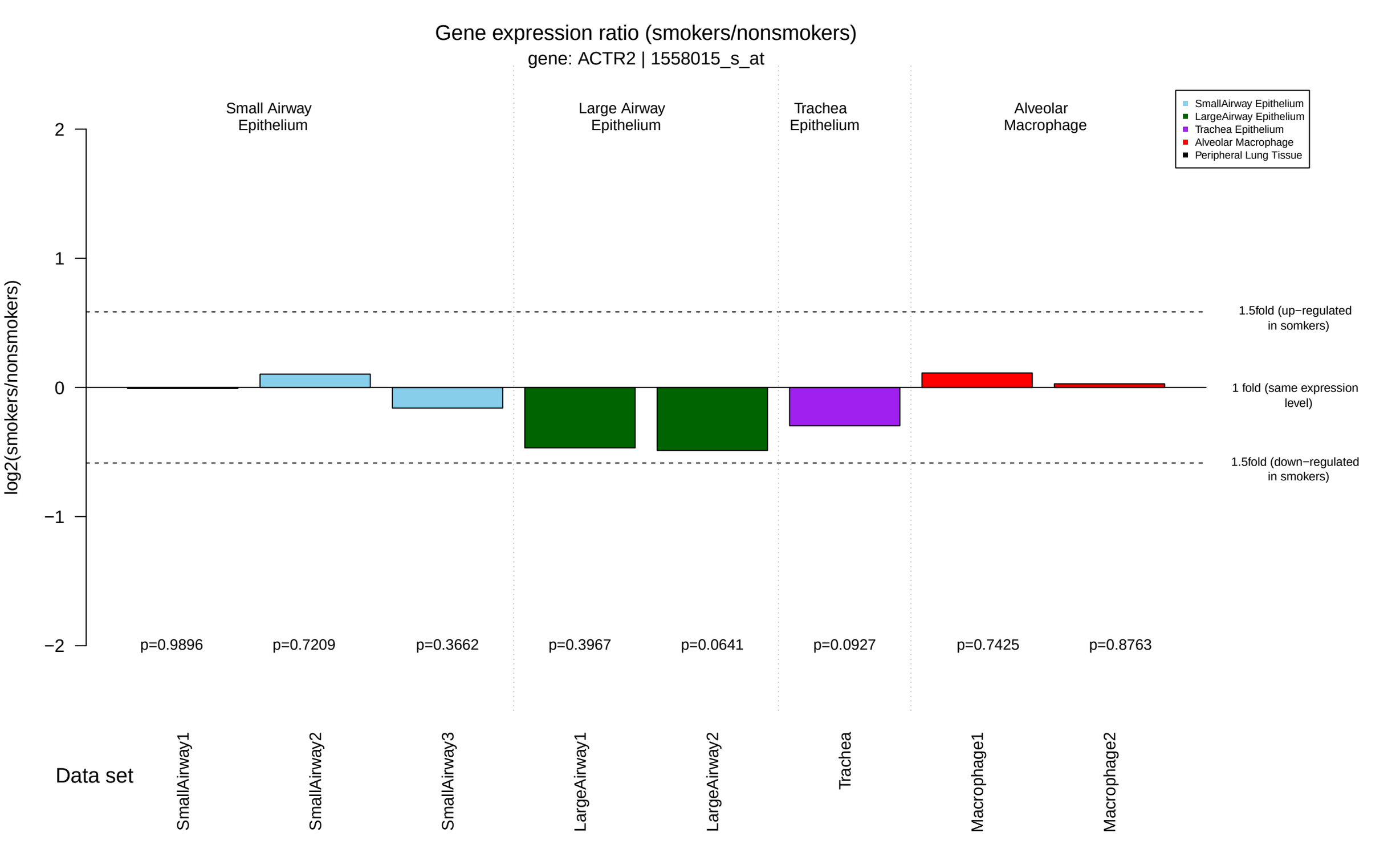
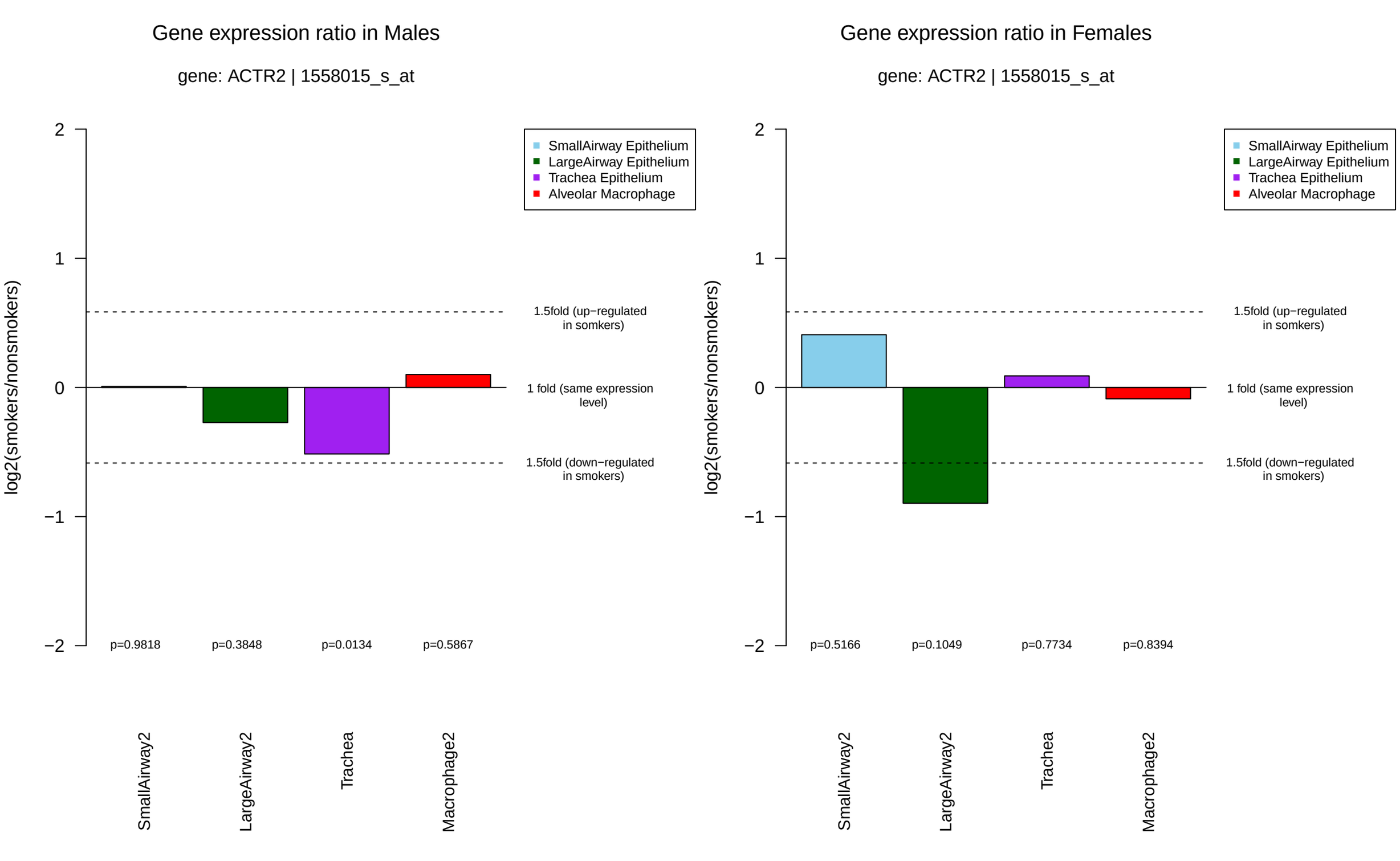
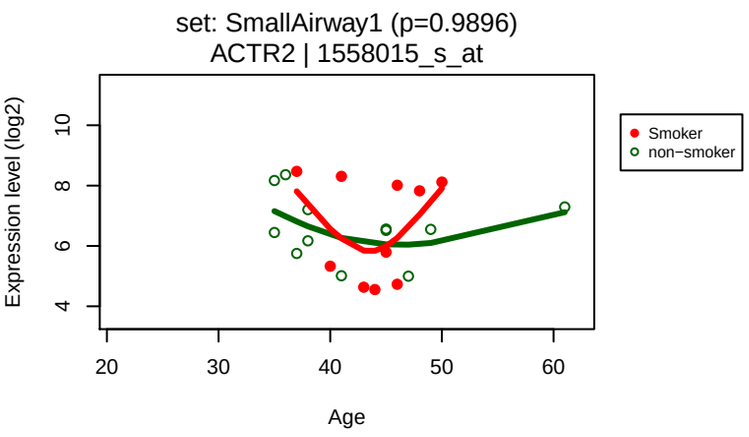
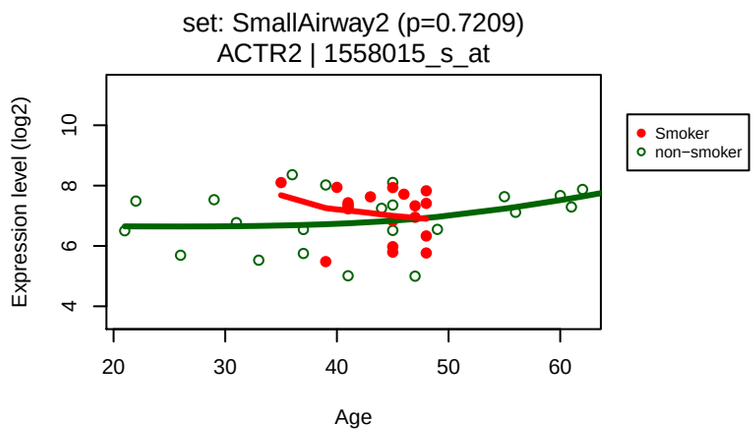
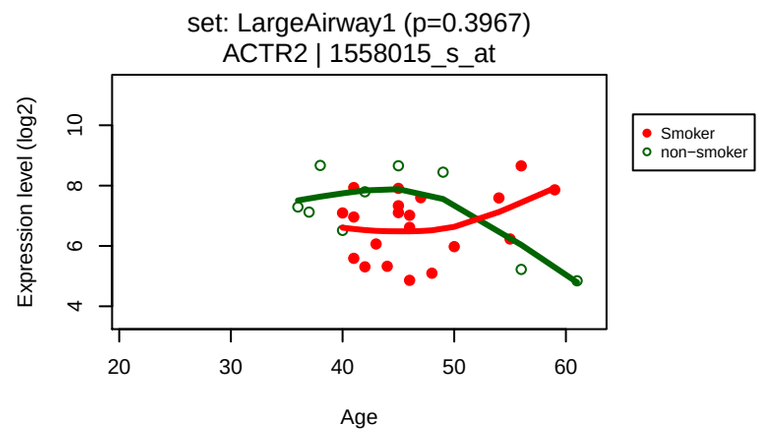

Gene expression of smokers and nonsmokers
gene: ACTR2 | 1558015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9896
p=0.7209
p=0.3662
p=0.3967
p=0.0641
p=0.0927
p=0.7425
p=0.8763
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

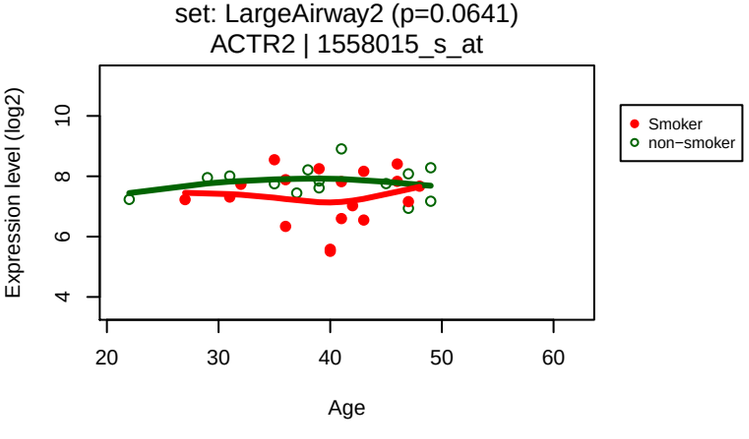

Gene expression ratio (smokers/nonsmokers)
gene: ACTR2 | 1558015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9896
p=0.7209
p=0.3662
p=0.3967
p=0.0641
p=0.0927
p=0.7425
p=0.8763
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

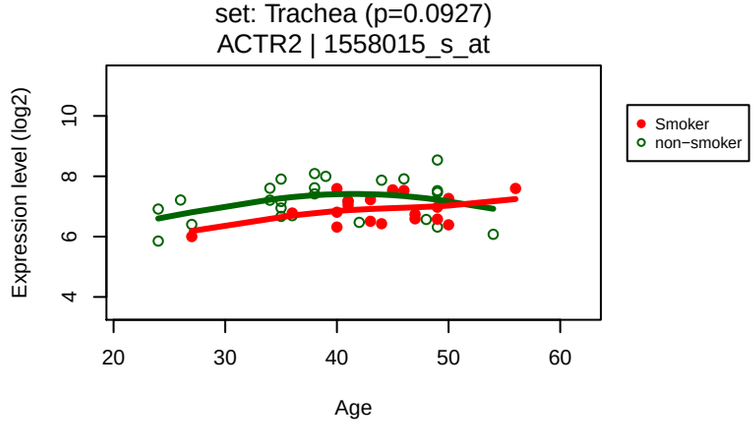

Gene expression ratio in Males
gene: ACTR2 | 1558015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9818
p=0.3848
p=0.0134
p=0.5867
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR2 | 1558015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5166
p=0.1049
p=0.7734
p=0.8394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

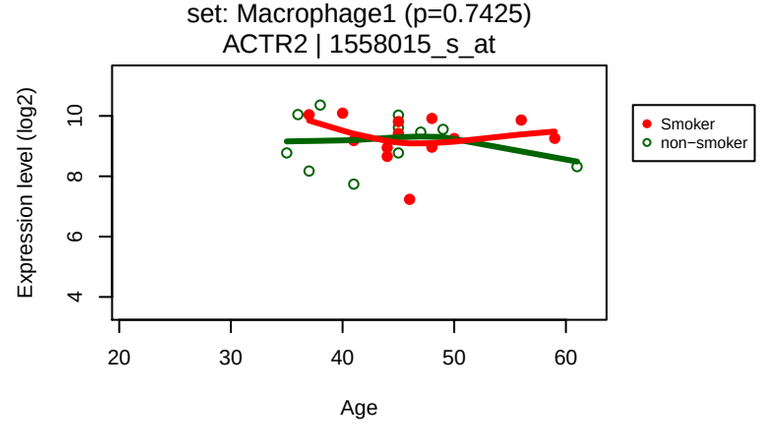

set: SmallAirway1 (p=0.9896)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker

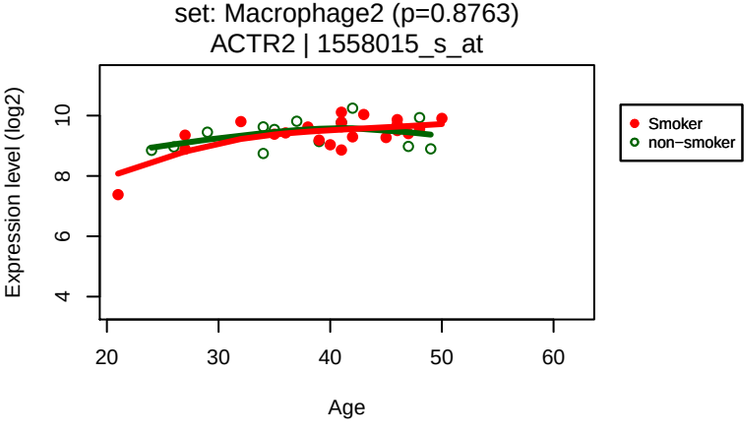

set: SmallAirway2 (p=0.7209)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker

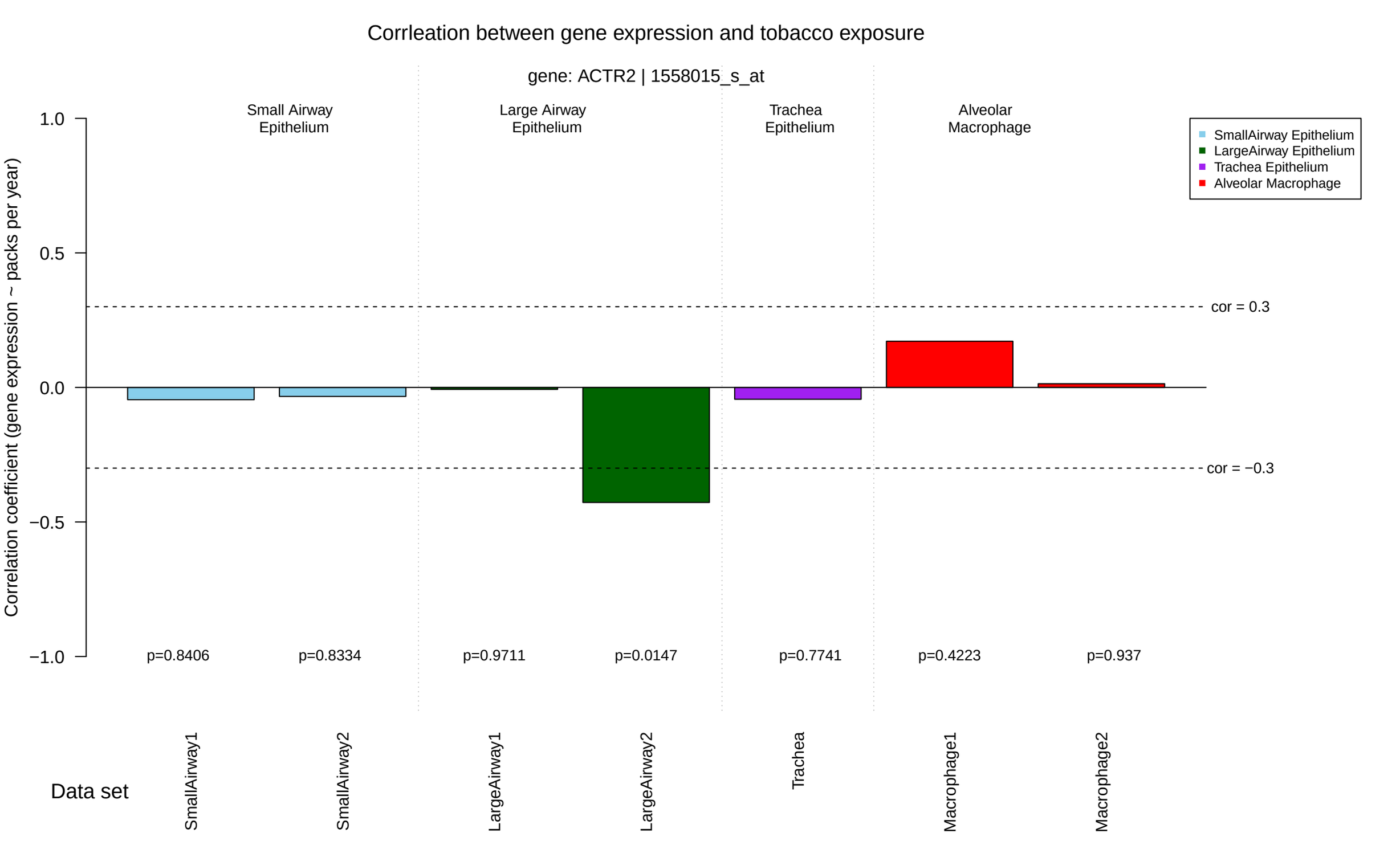

set: LargeAirway1 (p=0.3967)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0641)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0927)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7425)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8763)
ACTR2 | 1558015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR2 | 1558015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8406
p=0.8334
p=0.9711
p=0.0147
p=0.7741
p=0.4223
p=0.937
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2